

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	0.51%	0.48%	6.78%	0.82%
Meat MCal Target	9,291.44	8,836.29	123,786.47	15,035.81
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	132.54	0.00	1015.20	126.90
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	146.30	53.67	164.93	236.76
MCals from meat	9291.44	8836.29	123786.47	15035.81
Max Utilized Mcals allowed from Dairy	19391.15	0.00	167439.86	30044.30
MCals actually Utilized from Dairy	19391.15	0.00	167439.86	30044.30
Percent of diet from their meat	0.51%	0.48%	6.78%	0.82%
Percent of Diet from their Dairy	1.06%	0.00%	9.17%	1.65%
Kg Fleece	1,107.01	0.00	0.00	0.00
1 Year+ (age) skins	54.66	2.34	31.85	88.27
Yearly Offspring skins	266.77	63.83	243.26	431.33
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	261.26	9.80	242.55	422.78
Males Step 1	73.15	26.84	82.47	118.38
Females Step 1	73.15	26.84	82.47	118.38
Replacement Mothers per year	52.55	2.26	30.51	84.86
Breeding Males Required	13.72	0.51	12.73	22.20
Additional Mothers	215.12	1.86	115.19	347.45
Males Step 2	60.23	5.08	39.17	97.29
Females 2	60.23	5.08	39.17	97.29
Total Mothers	313.81	12.06	273.05	507.63
Total Males	133.38	31.92	121.63	215.66
Total Females	12.92	21.76	43.30	21.09
Total Offspring	266.77	63.83	243.26	431.33
Total Animals	473.83	66.25	450.72	766.59
Units per Hectare	1.00			
Total Adults per Hectare	2.06	0.32	0.51	2.06
Total Offspring per Hectare	1.02	1.39	0.31	1.01
	0.35%		4.27%	0.57%
Total Adults	543	14	401	877
Total Offspring	267	64	243	431
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	47.04	0.00	29.79	76.31
Other	4,268.31	0.00	0.00	6,901.22
MCals Dry Mothers	54,942.54	0.00	63,068.55	89,127.03
total Maintenance MCals	59,210.85	0.00	63,068.55	96,028.25

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	158,861	29,143	407,222	256,855
Lactating	303,076	10,360	437,873	490,030
Maintenance	59,211	0	63,069	96,028
Adult Males	16,020	0	26,957	25,925
Offspring	73,228	64,151	890,342	118,399
TOTAL	610,396.6	103,653.7	1,825,463.6	987,236.7

Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
	Crop Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	266.8	63.8	243.3	431.3
MCals from Straw/Chaff	34093.1	0.0	154077.1	55123.5
MCals from Grain	54020.7	12926.0	98521.5	87343.6
MCals from other feed	88113.8	12926.0	252598.6	142467.1

Diet Calculations		
Plants Mcal	1,425,624.686	
Meat Mcal	182,500	
Dairy	216,875.31365096	
	Perfect	
Plants %	78.12%	
Meat %	10.00%	
Dairy %	11.88%	
check	Perfect	
Percent		
Domestic Meat	8.60%	
Dairy	11.88%	
Wild Meat	1.40%	
Crops	70.30%	
Wild Plants	7.81%	
check	Perfect	
new dairy	216875.31	
potential dairy < limit	TRUE	
new limit	90.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	156,950.000	
Dairy	216,875.314	
Wild Meat	25550.000	
Crops	1,283,062.218	
Wild Plants	142,562.469	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd Males for Slaughter Females for Slaughter	314	12	273	508				
	133	32	122	216				
	13	22	43	21				
Offspring available for Consumption	267	64	243	431				
Gestation Req'd Lactation Req'd Maintenance Req'd Adult Males Offspring Required Energy Required	87124	24505	276098	140988				
	166216	8711	296879	268978				
	197986	0	81575	320174				
	16020	0	26957	25925				
	87634	69896	1065492	141691				
	554981	103113	1747001	897755				
Ewe Fleece Kg Yr ⁻¹	706							
Ram Fleece Kg year ⁻¹	27							
Juvenile Fleece Kg Yr ⁻¹	373							
Kg Fleece Year ⁻¹	1107	0	0	0				